Kainan Liu

AI Researcher & Engineer

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EDUCATION

Hong Kong University of Science and Technology (Guangzhou) 2023 – 2025
M.Sc. in Data-Centric Artificial Intelligence

Zhongnan University of Economics and Law 2019 – 2023
B.Sc. in Data Science and Big Data Technology

PUBLICATIONS

Astra: Activation-Space Tail-Eigenvector Low-Rank Adaptation of Large Language Models

ACL 2026 Findings

†Kainan Liu*, Yong Zhang, Ning Cheng, Yun Zhu, Yanmeng Wang, Shaojun Wang, Jing Xiao

Proposes a novel PEFT method that leverages tail eigenvectors of the activation covariance matrix — estimated from a task-specific calibration set — to construct task-adaptive low-rank adapters. Achieves faster convergence and improved downstream performance with significantly fewer parameters compared to standard LoRA approaches.

GRASP: Replace Redundant Layers with Adaptive Singular Parameters for Efficient Model Compression

EMNLP 2025 Main Conference

†Kainan Liu*, Yong Zhang, Ning Cheng, Zhitao Li, Shaojun Wang, Jing Xiao

A gradient-guided compression framework that identifies functionally redundant transformer layers and replaces them with learnable singular value parameters. Preserves model performance while substantially reducing parameters and inference cost, enabling efficient deployment of LLMs.

Detecting and dissecting anomalous anatomic regions in spatial transcriptomics with STANDS

Nature Communications

Kaichen Xu†, Yan Lu, Suyang Hou, Kainan Liu, Yihang Du, Mengqian Huang, Hao Feng, Hao Wu, Xiaobo Sun*

A GAN-based multi-task deep learning framework that integrates spatial information and gene expression profiles to detect and dissect anomalous tissue domains in spatial transcriptomics data at unprecedented resolution.

† First author * Corresponding author

EXPERIENCE

Ping An Technology (Shenzhen) Co., Ltd. Aug 2025 – Present
AI Researcher & Engineer

- Training and optimizing LLMs for financial risk control and corporate banking analysis, including domain-adaptive fine-tuning and prompt optimization for domain-specific tasks
- Developed Astra (ACL 2026 Findings): an activation-space low-rank adaptation method for efficient LLM fine-tuning in financial domains
- Building evaluation and monitoring pipelines to track model performance across financial NLP benchmarks

Ping An Technology (Shenzhen) Co., Ltd. Sept 2024 – Aug 2025
Research Intern

- Conducted research on model compression techniques, focusing on identifying and replacing redundant transformer layers with parameter-efficient alternatives
- Developed GRASP (EMNLP 2025 Main Conference), a gradient-guided compression framework for large language models
- Implemented and benchmarked various pruning and compression methods on models up to 7B parameters

Vipshop (China) June 2024 – Sept 2024
Recommendation Algorithm Intern

- Processed and analyzed large-scale user-item interaction data and behavioral sequences for recommendation system enhancement
- Deployed XGBoost models for recommendation ranking optimization, improving key engagement metrics
- Developed feature engineering pipelines for extracting user behavior patterns from historical interaction data

TECHNICAL SKILLS

Programming	Python (primary), PyTorch, NumPy, Pandas
Frameworks	PyTorch, PEFT, Transformers, vLLM, XGBoost, scikit-learn
Research Areas	Low-rank adaptation (LoRA, DoRA, Astra), Model compression, AI Agents, Agent data synthesis, LLM evaluation
Languages	Chinese (native), English (professional proficiency)